Compare the SOC analyst, VAPT/penetration tester and incident responder roles in a table: goal, daily work, tools, blue/red team.

Cybersecurity has different roles that work together to protect computer systems, networks, and digital information. Three important roles are the Security Operations Center (SOC) Analyst, the Vulnerability Assessment and Penetration Testing (VAPT) professional, and the Incident Responder.

Although all three roles aim to improve security, their responsibilities and approaches differ.

1. SOC Analyst

A **Security Operations Center (SOC) Analyst** monitors an organization's systems and networks to detect, investigate, and report suspicious activities.

Main responsibilities:

- Monitor security alerts and system logs.
- Investigate suspicious network traffic and login attempts.
- Identify possible malware infections and unauthorized access.
- Use security tools to detect threats.
- Escalate serious incidents to the appropriate team.

Common tools: SIEM platforms, EDR tools, Wireshark, and log-analysis tools.

Team: Primarily Blue Team (defensive security).

Example: A SOC analyst investigates repeated failed login attempts followed by a suspicious successful login.

2. VAPT / Penetration Tester

A **Vulnerability Assessment and Penetration Testing (VAPT) professional** identifies security weaknesses in systems, networks, and applications. A penetration tester performs authorized testing to determine whether identified weaknesses can be exploited and what their potential impact is.

Main responsibilities:

- Identify vulnerabilities in authorized targets.
- Scan systems and applications for weaknesses.
- Validate security findings within the agreed scope.

- Document evidence and assess potential impact.
- Recommend fixes and prepare a security report.

Common tools: Nmap, Burp Suite, OWASP ZAP, and Wireshark.

Team: Primarily Red Team (offensive security testing).

Example: A penetration tester discovers that a test web application has an access-control weakness and documents how it could expose another user's data.

3. Incident Responder

An **Incident Responder** investigates and manages security incidents to limit damage, remove threats, and restore affected systems.

Main responsibilities:

- Investigate confirmed or suspected security incidents.
- Determine which systems and information are affected.
- Contain threats, such as isolating an infected computer.
- Preserve evidence for further investigation.
- Support recovery and recommend preventive improvements.

Common tools: EDR platforms, SIEM tools, forensic tools, and network-analysis tools.

Team: Primarily Blue Team (defensive security and incident handling).

Example: An incident responder investigates a ransomware infection, isolates affected systems, determines the scope of the incident, and helps restore systems safely.

Category	SOC Analyst	VAPT / Penetration Tester	Incident Responder
Goal	Detect and investigate suspicious activity.	Find and validate security weaknesses.	Contain, investigate, and recover from security incidents.
Daily work	Review alerts, analyze logs, investigate threats, and escalate cases.	Scan authorized targets, test weaknesses, and prepare reports.	Investigate incidents, preserve evidence, contain threats, and support recovery.
Tools	SIEM, EDR, Wireshark, log-analysis tools.	Nmap, Burp Suite, OWASP ZAP, Wireshark.	EDR, SIEM, forensic tools, packet-analysis tools.
Team	Primarily Blue Team.	Primarily Red Team.	Primarily Blue Team.

Approach	Continuous monitoring and detection.	Planned, authorized security testing.	Incident-driven investigation and recovery.
Main output	Alerts, investigations, and incident escalations.	Vulnerability findings and remediation recommendations.	Incident reports, evidence, and recovery actions.

Blue Team vs Red Team

- **Blue Team:** Focuses on defending systems, monitoring activity, detecting attacks, and responding to threats.
- **Red Team:** Simulates adversarial techniques within an authorized scope to evaluate an organization's defenses.
- **Purple Team:** Encourages collaboration between offensive and defensive teams to improve detection and response.

SOC analysts and incident responders generally work on the defensive side. Penetration testers generally perform authorized offensive testing. However, job responsibilities can overlap, and not every penetration test is a full Red Team engagement.

Conclusion

SOC analysts, penetration testers, and incident responders have different but complementary responsibilities.

- A **SOC Analyst** detects and investigates suspicious activity.
- A **VAPT/Penetration Tester** identifies and validates security weaknesses.
- An **Incident Responder** investigates security incidents, contains threats, and supports recovery.

Together, these roles help organizations identify weaknesses, detect attacks, and protect their systems and information.